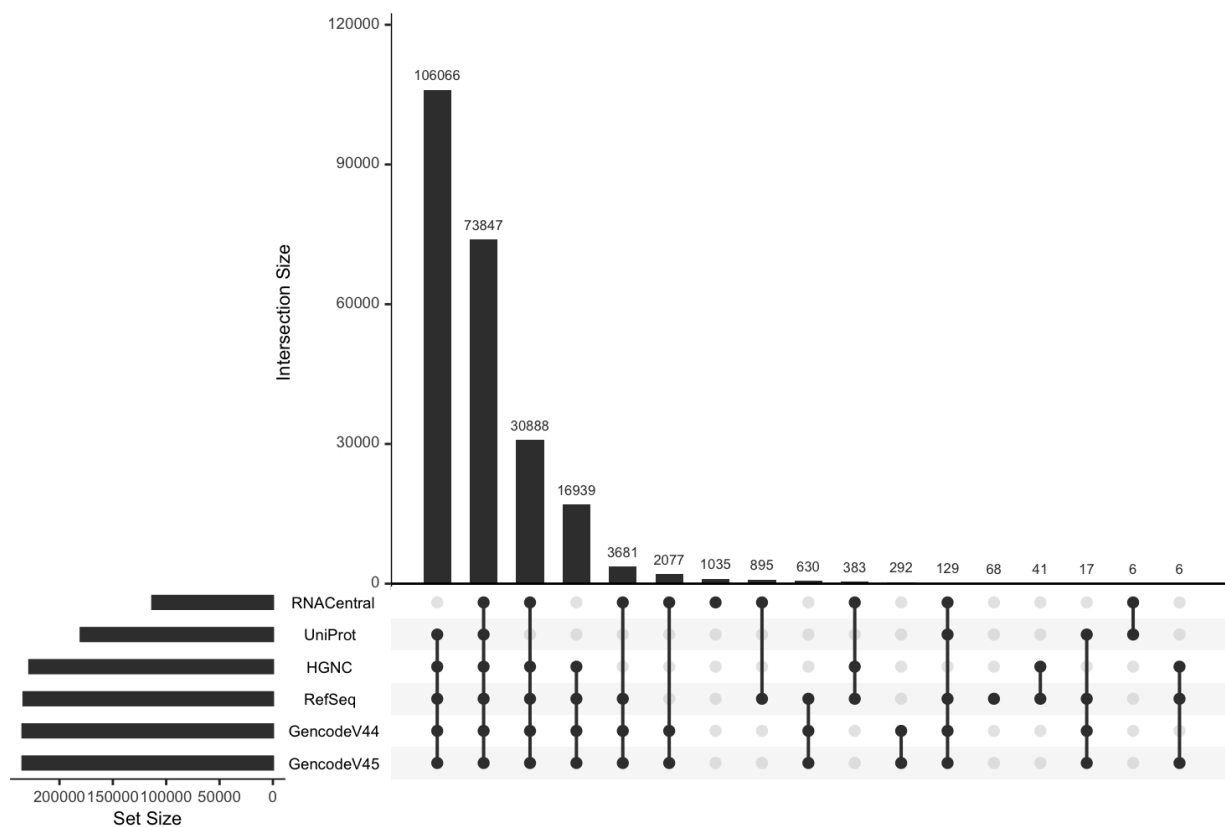
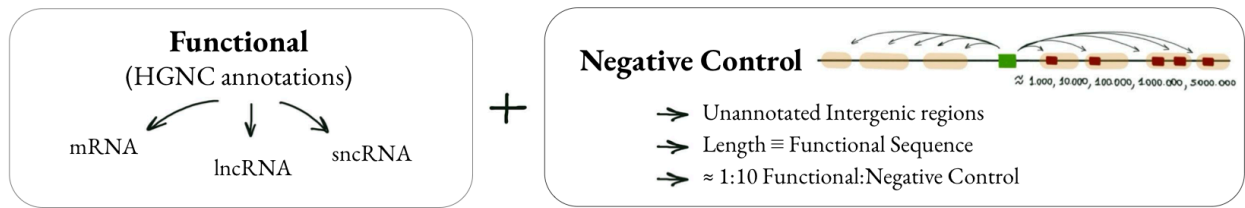


Supplementary Figures

We generated an Upset plot to determine which genome annotation databases should be used to select negative control sequences that corresponded to intergenic unannotated regions of the genome. The R function ‘upset’ from the UpSetR v1.4.0 package was used. The used script (“UpsetPlot.R”) is included in the project repository. The databases that contained the highest number of annotations (GencodeV45) and unique annotations (RNACentral) were selected, as they encapsulated the most annotations.



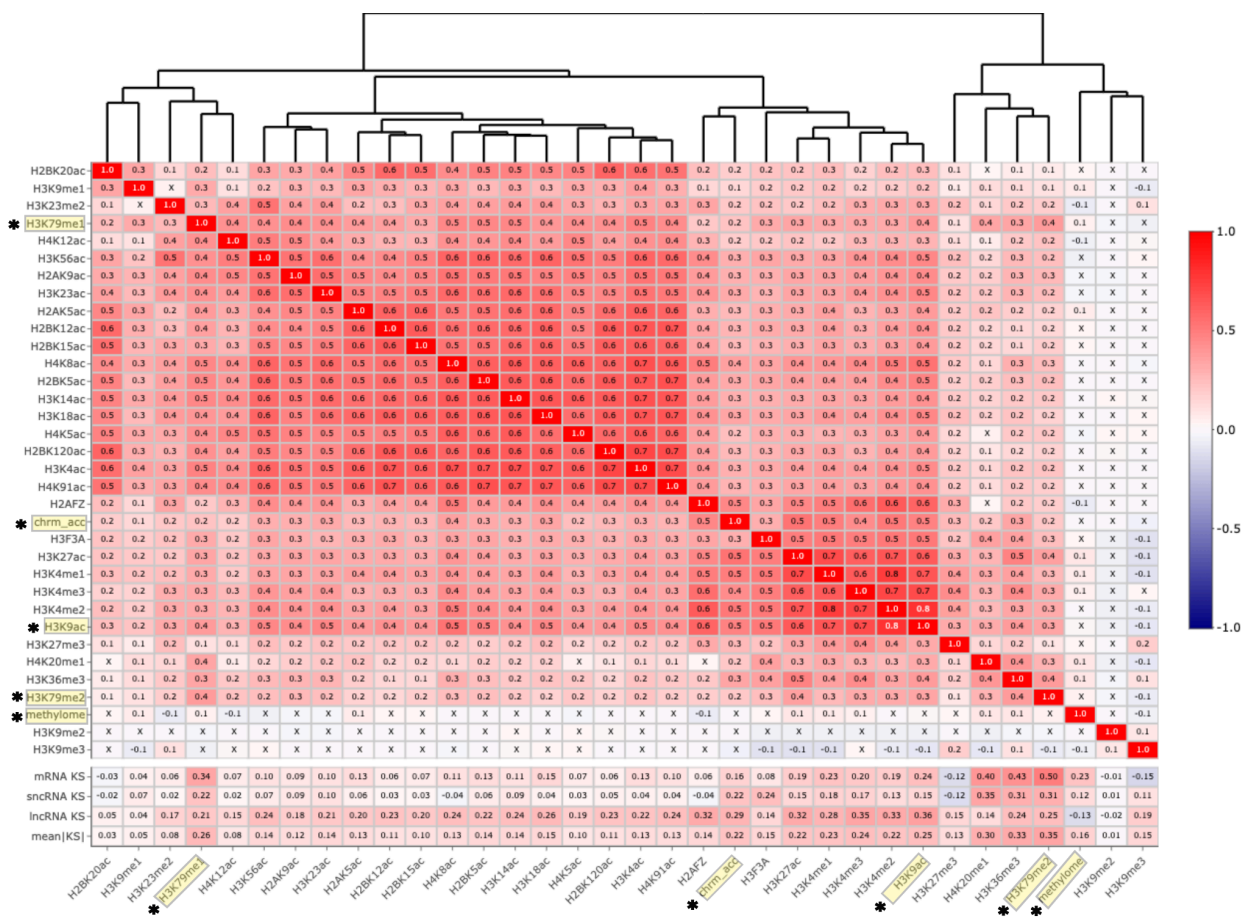
Supplementary Figure S1. Upset plot for different human genome databases (build GRCh38) corresponding to annotations from chromosome 22. This figure illustrates the overlap of gene annotations from six prominent databases: RNACentral, UniProt, HGNC, RefSeq, Gencode V44, and Gencode V45. The horizontal bars represent the total number of gene annotations in each database. The vertical bars represent the number of gene annotations that are common (or unique) across the selected combinations of databases, indicated by the connected dots below. The connected dots indicate the specific databases involved in each intersection.



Supplementary Figure S2. Workflow for generating control datasets. Functional sequences were selected from HGNC-linked mRNA, lncRNA, and sncRNA annotations. For each functional sequence, length-matched negative controls were sampled from unannotated intergenic regions of the GRCh38 genome at increasing distances from the corresponding gene (approximately 1, 10, 100, 1000, and 5000 kb upstream and downstream). Candidate controls were filtered to remove overlaps with known annotations, generating a final set of negative controls at an approximate 1:10 functional-to-control ratio.

Selection of most relevant epigenetic features

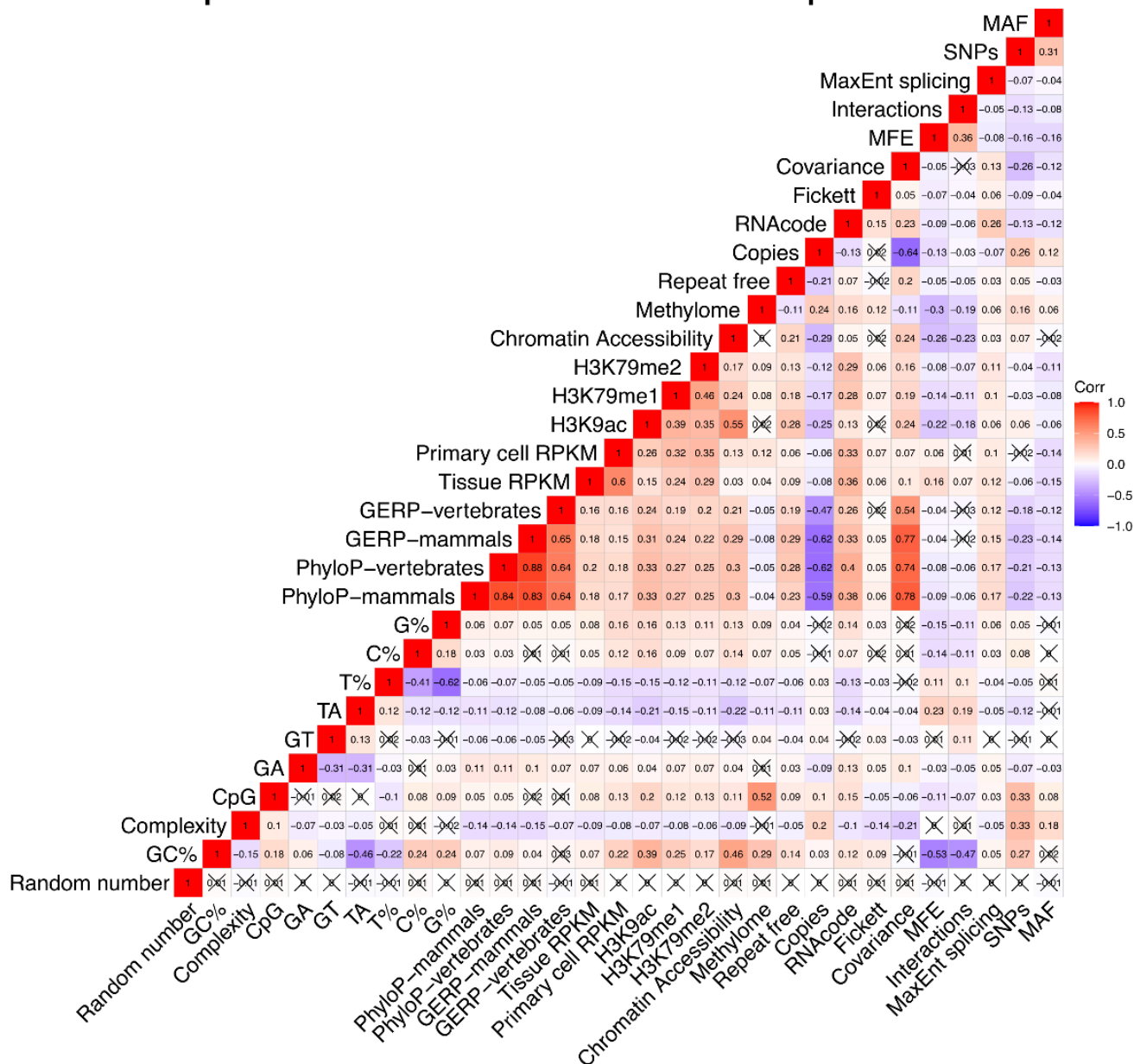
A Spearman correlation matrix was generated to assess correlations among epigenetic features and identify the most informative and uncorrelated variables for subsequent analysis. Kolmogorov-Smirnov statistics were calculated to quantify distributional differences between positive and negative control sequences, enabling identification of features with the greatest discriminatory power. These plots were created using the R functions 'rcorr' from Hmisc v.5.2-3 package, 'heatmaply' from Heatmaply v1.5.0 package, 'p.adjust' and 'ks.test' from Stats v.4.4.1 package



Supplementary Figure S3. Correlation matrix of epigenetic features. Spearman correlation of histone and DNA modifications for mRNA, snRNA and lncRNA sequences. Positively correlated features are displayed in red and negatively correlated features in blue. Spearman's Rho was used to determine the significance of each correlation, which were corrected for multiple testing using a Benjamini-Hochberg (BH) correction. Non-significant correlations with a p-value > 0.05 have been marked with a cross. At the bottom of the plot, KS statistics are included for each gene type as well as a mean |KS| for overall comparison.

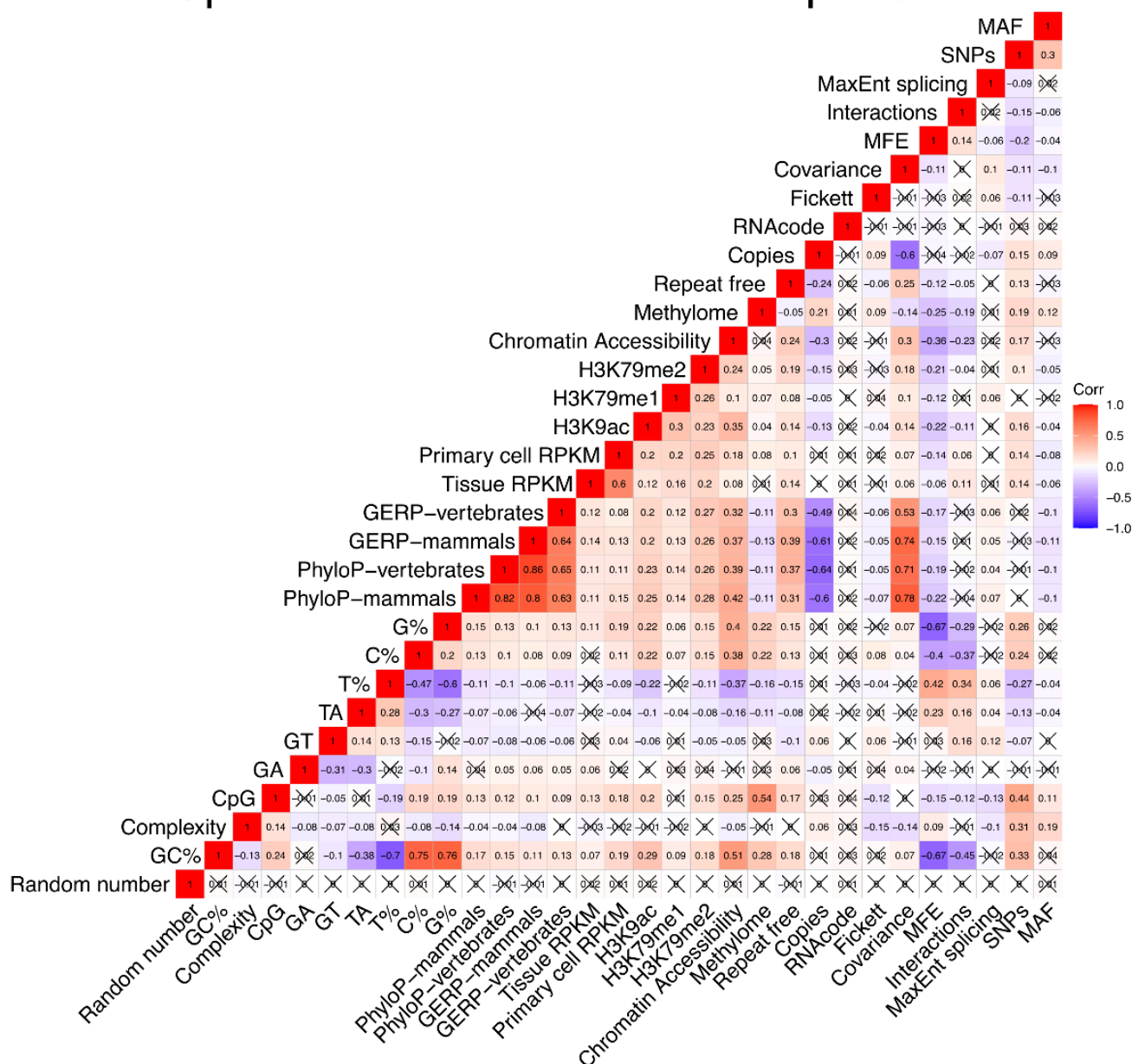
A Spearman correlation matrix was generated to assess correlation among selected features of gene functionality. These plots were created using the R functions 'ggcorrplot' from ggcorrplot v.0.1.4.1 package and 'p.adjust' from Stats v.4.4.1 package

Spearman correlation Heatmap – mRNA



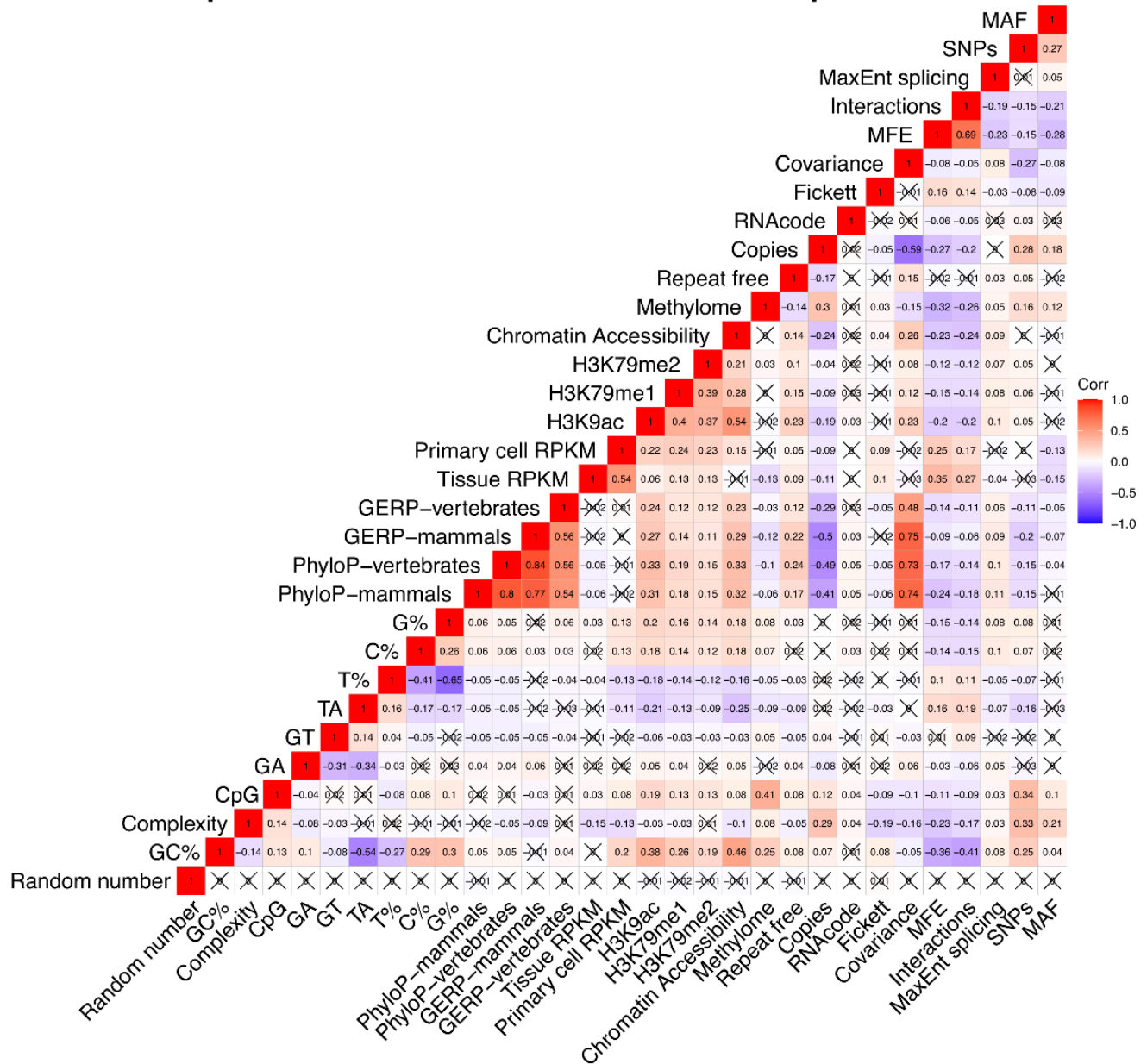
Supplementary Figure S4. mRNA correlation matrix. Spearman correlation of functionality predictors for protein-coding sequences. Positively correlated features are displayed in red and negatively correlated features in blue. Spearman's Rho was used to determine the significance of each correlation, which were corrected for multiple testing using a Bonferroni correction. Non-significant correlations with a p-value > 0.05 have been marked with a cross.

Spearman correlation Heatmap – sncRNA



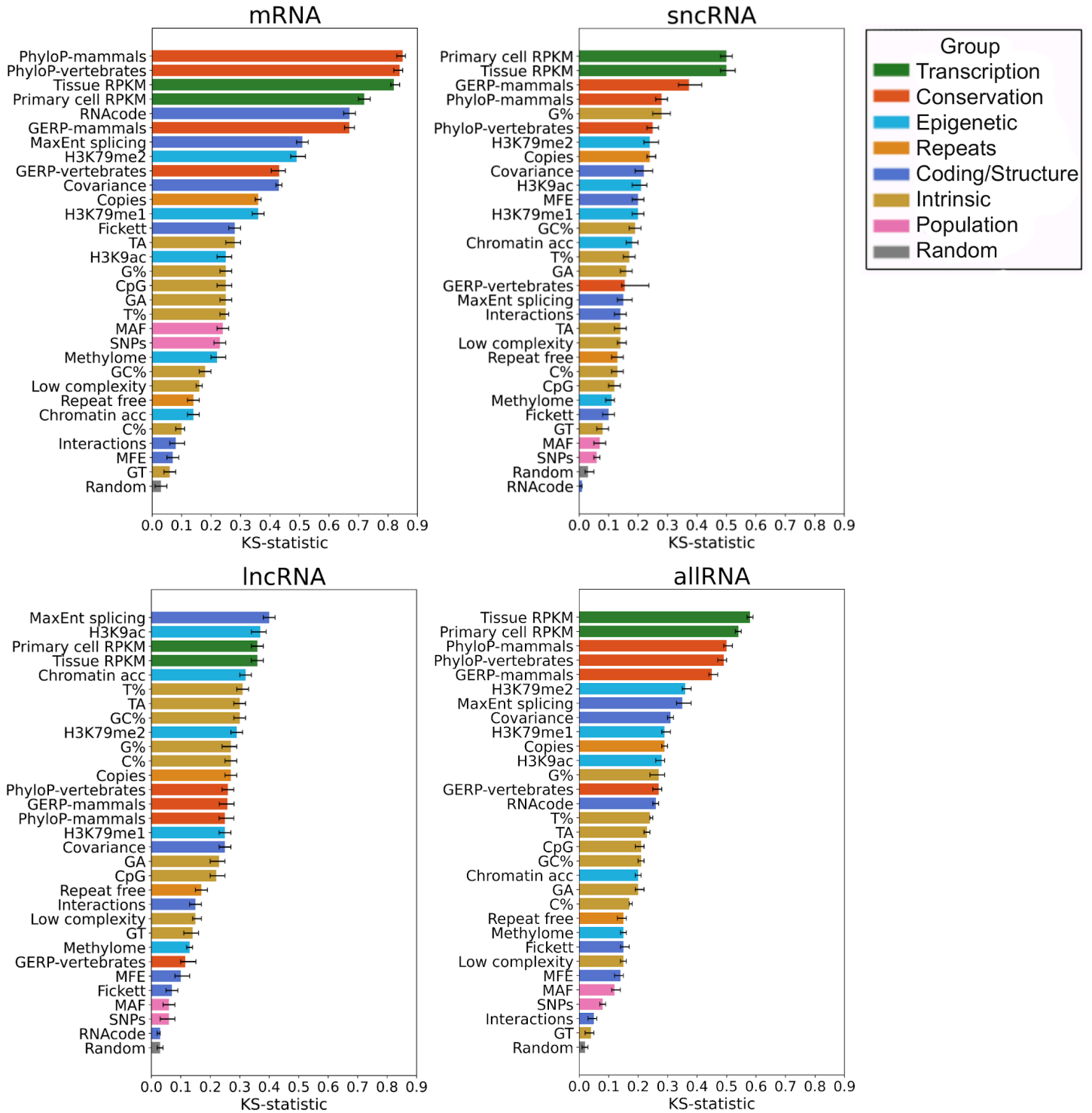
Supplementary Figure S5. sncRNA correlation matrix. Spearman correlation of functionality predictors for sncRNA sequences. Positively correlated features are displayed in red and negatively correlated features in blue. Spearman's Rho was used to determine the significance of each correlation, which were corrected for multiple testing using a Bonferroni correction. Non-significant correlations with a p-value > 0.05 have been marked with a cross.

Spearman correlation Heatmap – IncRNA



Supplementary Figure S6. IncRNA correlation matrix. Spearman correlation of functionality predictors for IncRNA sequences. Positively correlated features are displayed in red and negatively correlated features in blue. Spearman's Rho was used to determine the significance of each correlation, which were corrected for multiple testing using a Bonferroni correction. Non-significant correlations with a p-value > 0.05 have been marked with a cross.

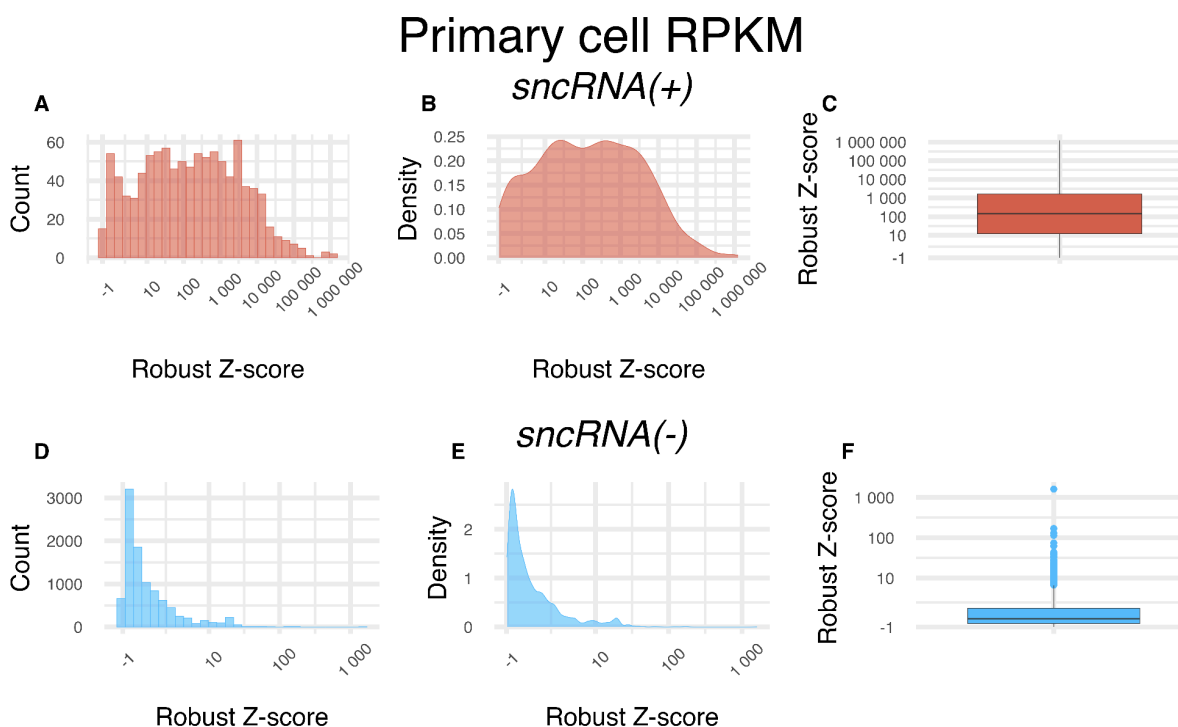
Data effect sizes: KS-statistic by dataset



Supplementary Figure S7. Data effect sizes. Features ranked by Kolmogorov-Smirnov (KS) statistic. Each point includes 95% confidence intervals, calculated using bootstrapping (1000 replicates). Different colors represent feature classes.

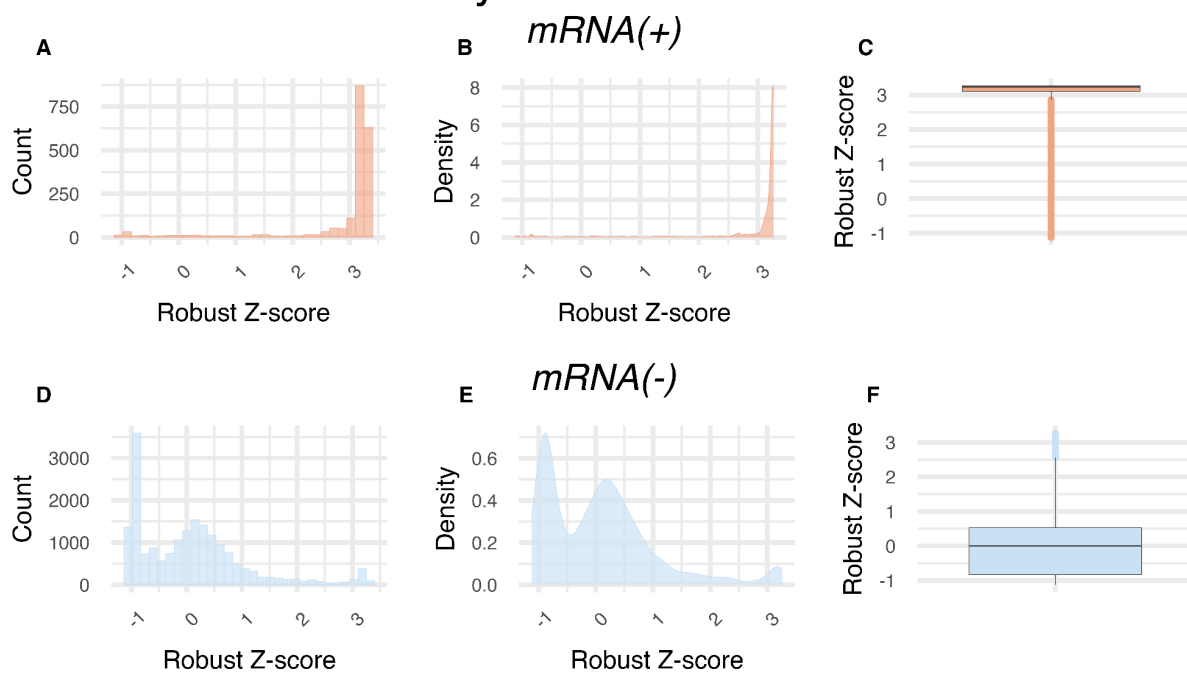
Genome feature example distributions summaries

For some of the genome features we illustrate example distributions that are used to generate the robust z-scores violin plots shown in the manuscript. Several features exhibit highly skewed distributions (e.g., primary cell RPKM), which explains the substantial variation in z-scores observed between the positive and negative gene sets. These plots were created using the R functions 'geom_histogram', 'geom_density', 'geom_boxplot' and 'stat_qq' from the ggplot2 v3.5.1 package. The distributions for each feature and class can be found in the GitHub repository ([results/Paper figures/Distribution_plots](#)).

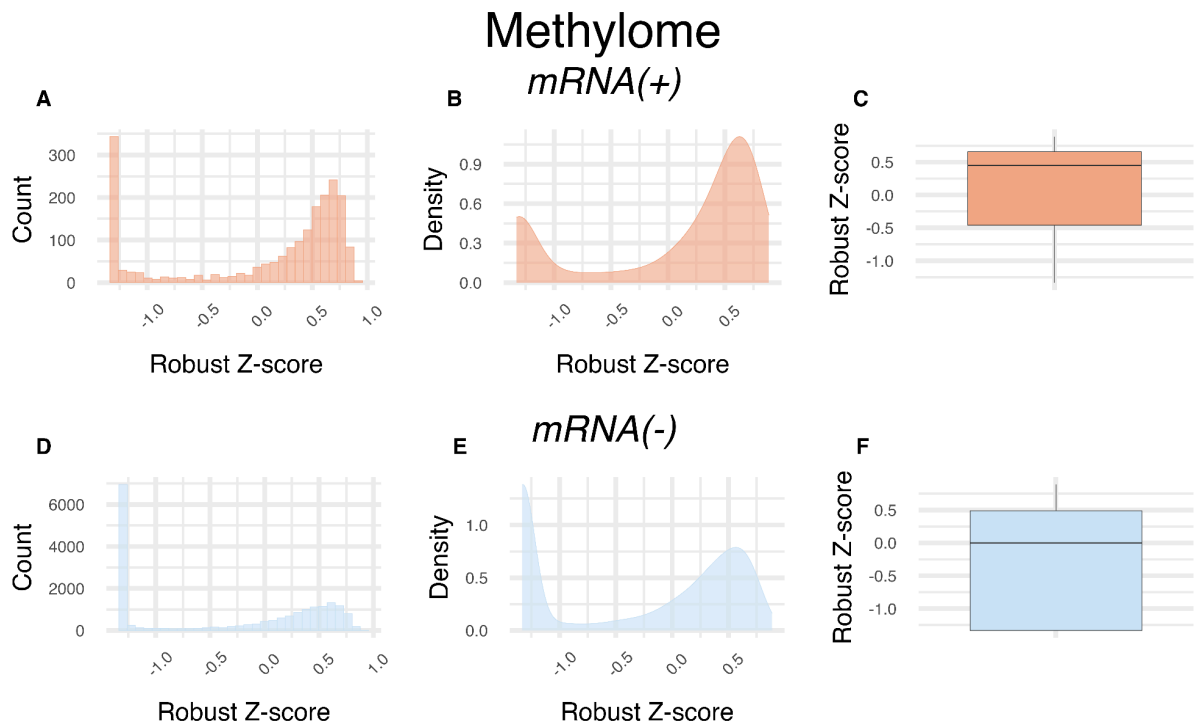


Supplementary Figure S8. *sncRNA* primary cell RPKM robust z-scores distributions in log scale. A-C) Histogram, Density and Boxplot for *sncRNA(+)*, D-F) Histogram, Density and Boxplot for *sncRNA(-)* all show the robust z-scores for maximum primary cell RPKM values.

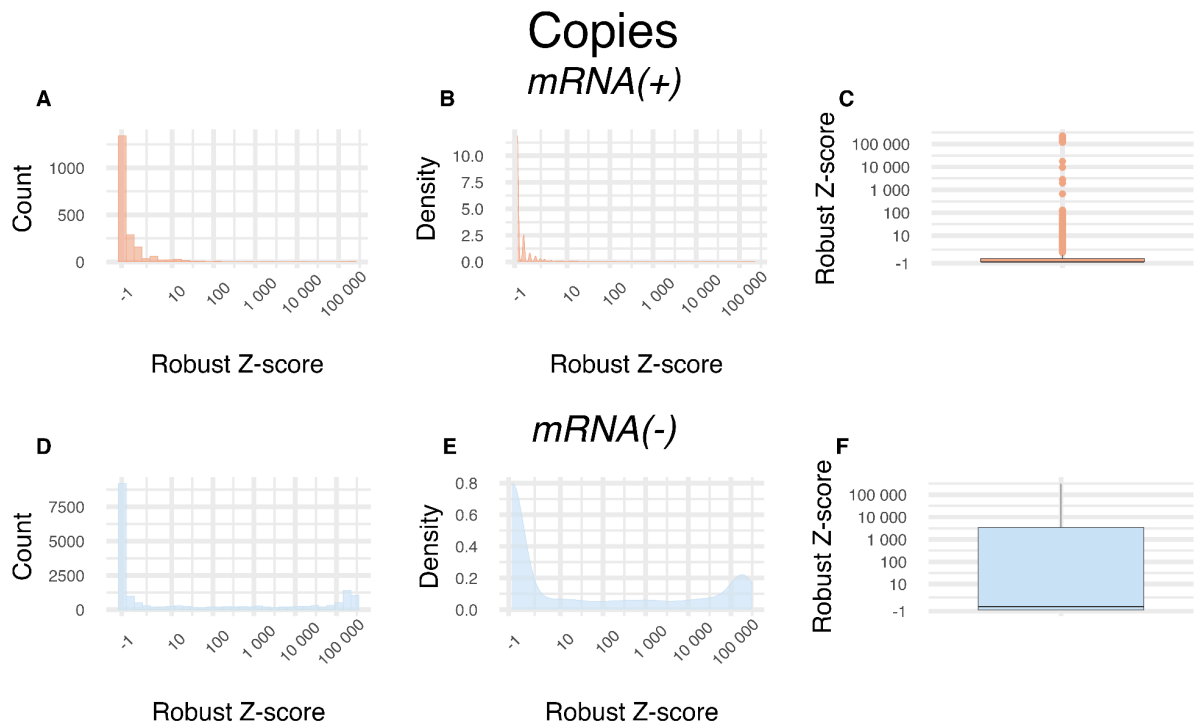
PhyloP mammals



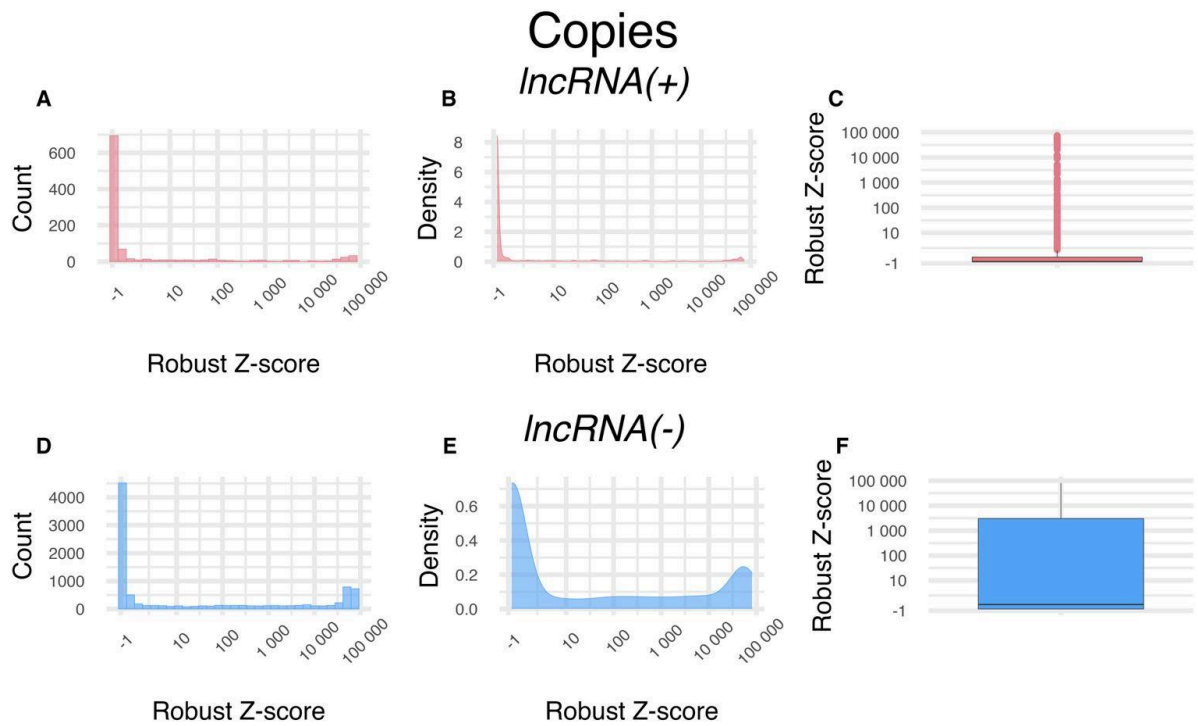
Supplementary Figure S9. mRNA PhyloP mammals robust z-scores distributions. A-C) Histogram, Density and Boxplot for mRNA(+), D-F) Histogram, Density and Boxplot for mRNA(-) for robust z-scores generated from PhyloP mammal conservation values.



Supplementary Figure S10. mRNA methylome robust z-scores distributions in log scale. A-C) Histogram, Density and Boxplot for mRNA(+), D-F) Histogram, Density and Boxplot for mRNA(-) robust z-scores computed from the average proportion of methylated CpG sites across samples.



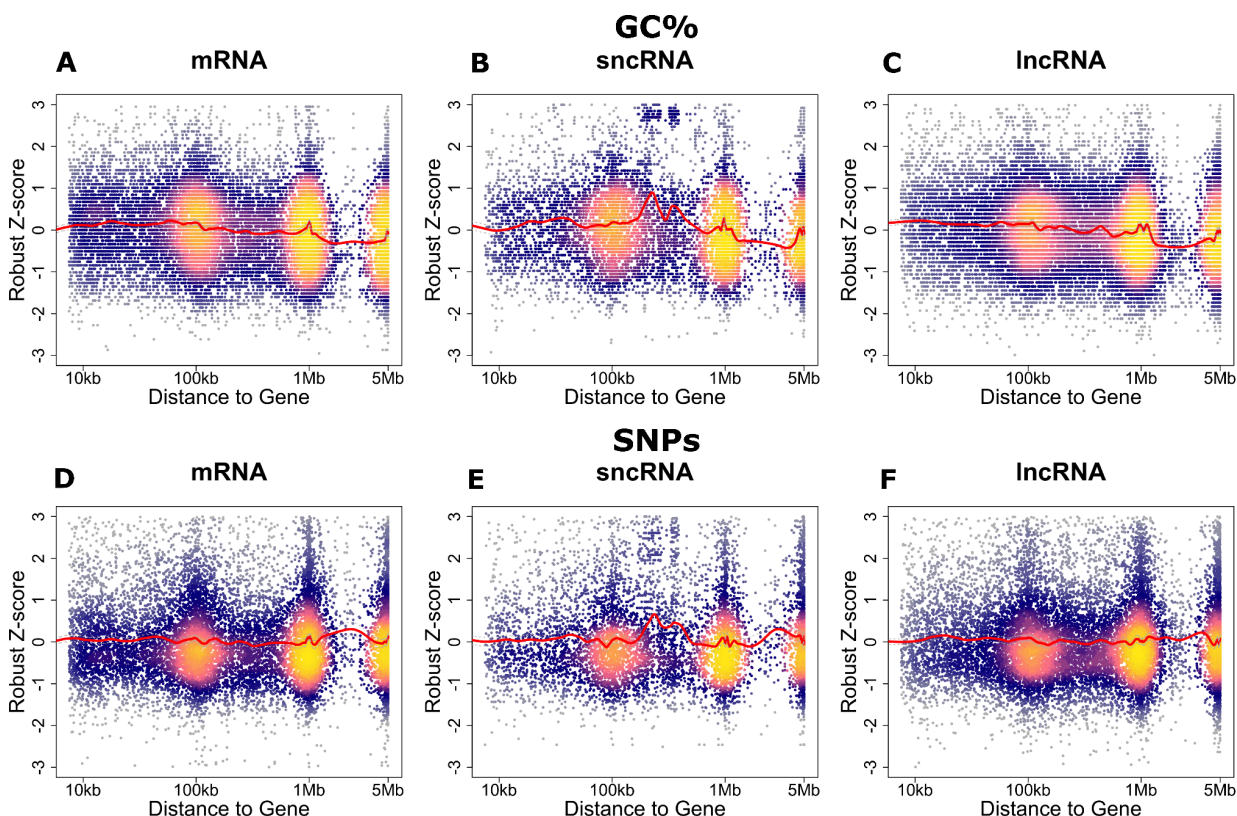
Supplementary Figure S11. mRNA copies robust z-scores distributions in log scale. A-C) Histogram, Density and Boxplot for mRNA(+), D-F) Histogram, Density and Boxplot for mRNA(-) of z-scores computed from the number of genomic copies of each region.



Supplementary Figure S12. lncRNA copies robust z-scores distributions in log scale. A-C) Histogram, Density and Boxplot for *lncRNA(+)*, D-F) Histogram, Density and Boxplot for *lncRNA(-)* of z-scores computed from the number of genomic copies of each region.

Effect on distance

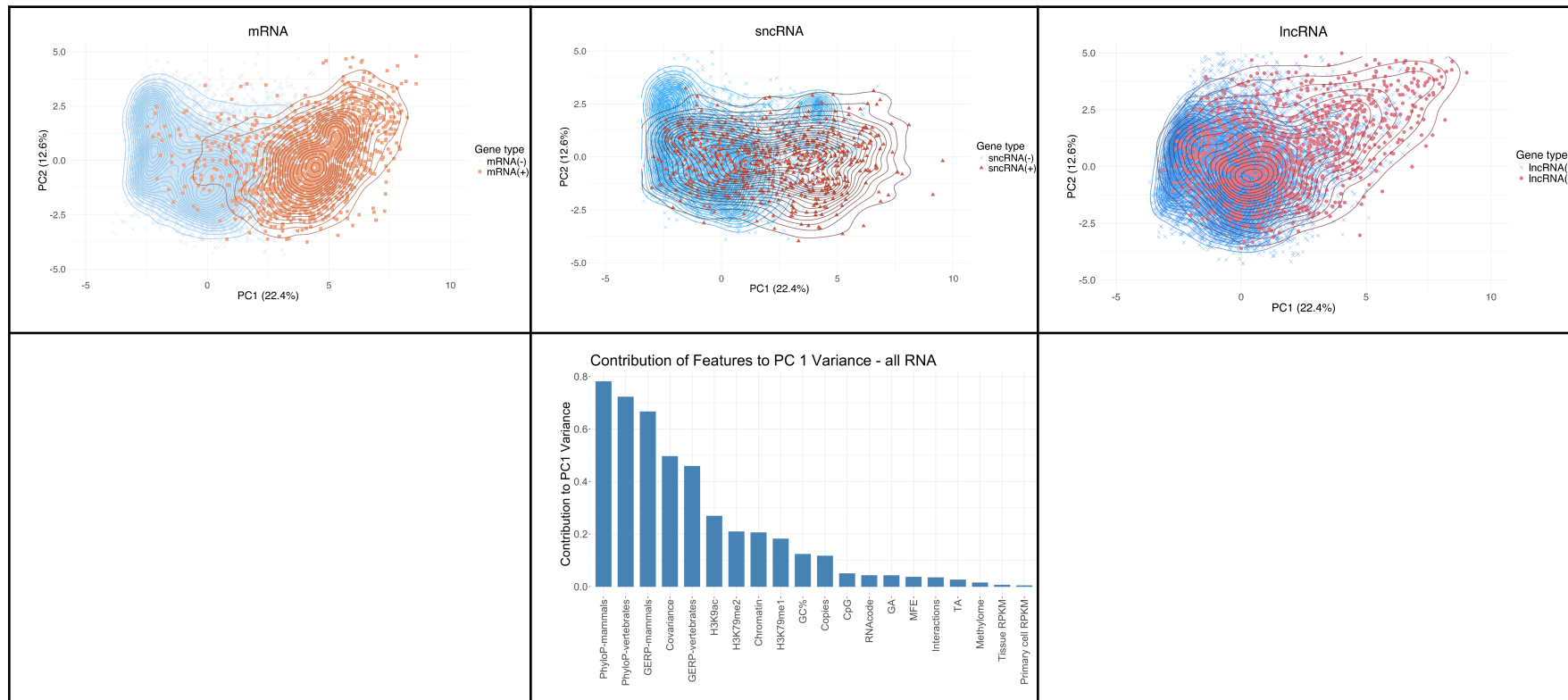
In order to demonstrate that there was no effect on distance to gene for the negative control sequences, Spearman correlation coefficients were calculated between robust z-scores using the R functions `rcorr` from Hmisc v. 5.2-3 package. Then to create the heatscatter plots the function `heatscatter` from LSD v. 4.1-0 package was used, and to compute the trend line the `loess` function from stats v. 4.4.1 package was used.

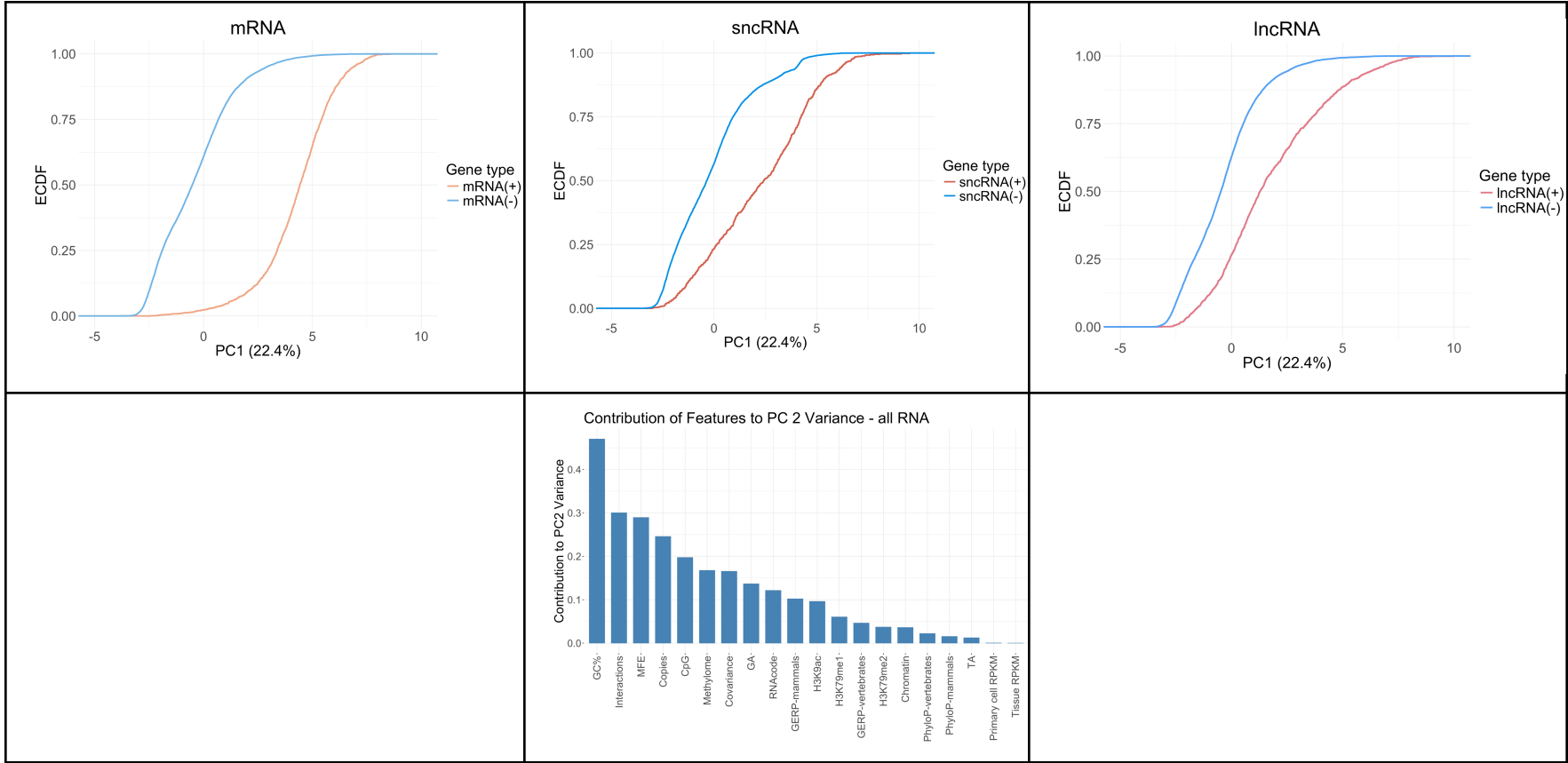


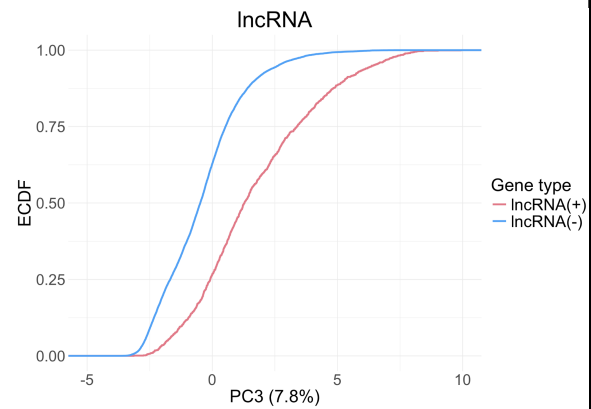
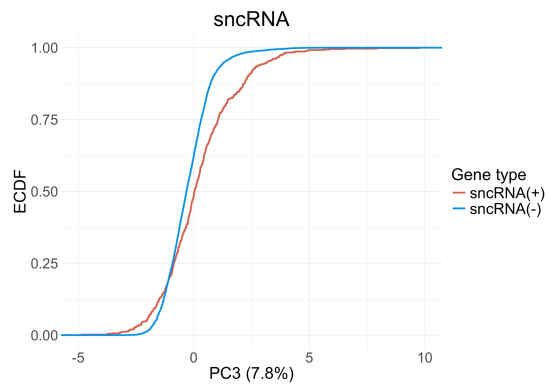
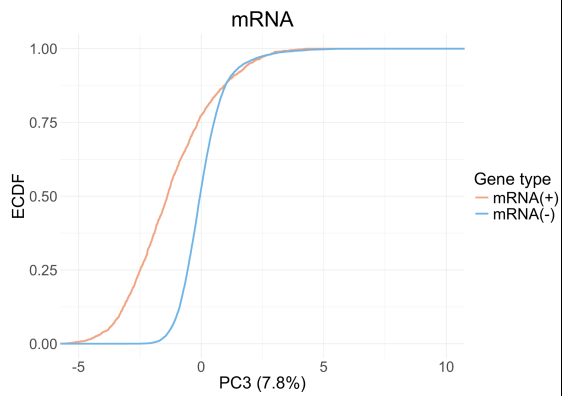
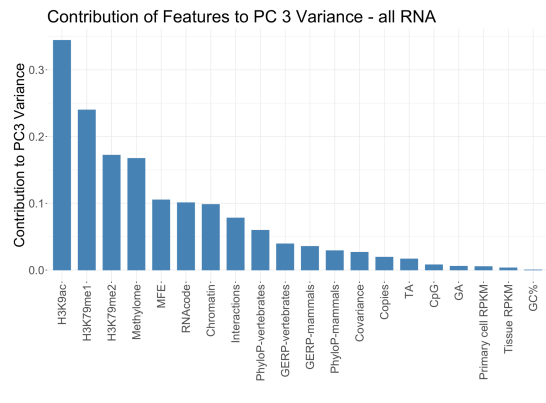
Supplementary Figure S13. Heatscatter for the effect on distance. A-C) GC% for each gene type, D-F) SNPs for each gene type. Robust Z-scores are displayed on the y-axis as a function of the distance to the nearest gene in the x-axis. Each point represents a sequence. There is no clear linear relationship between distance to the gene and GC content variation or SNPs.

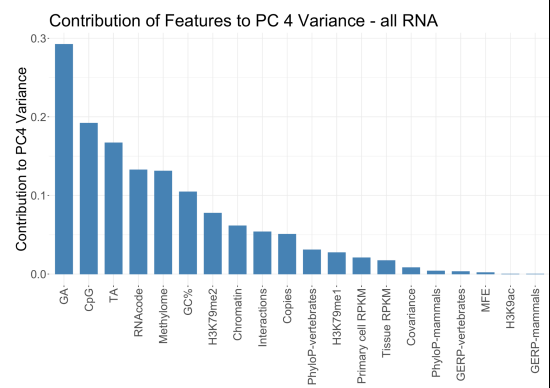
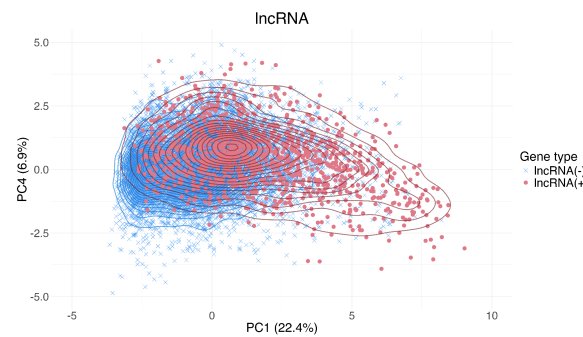
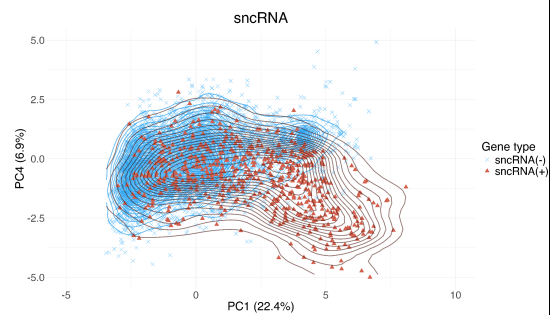
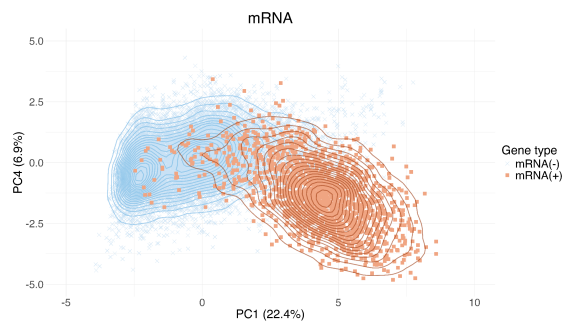
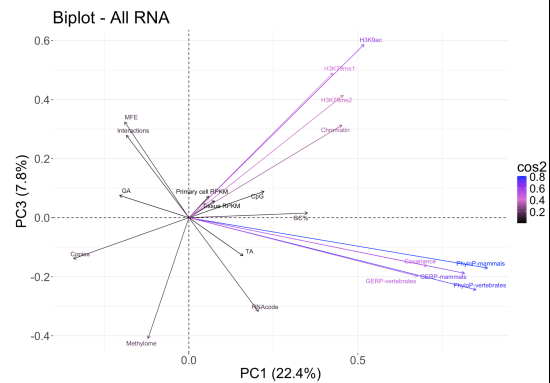
Result summaries/Data overviews

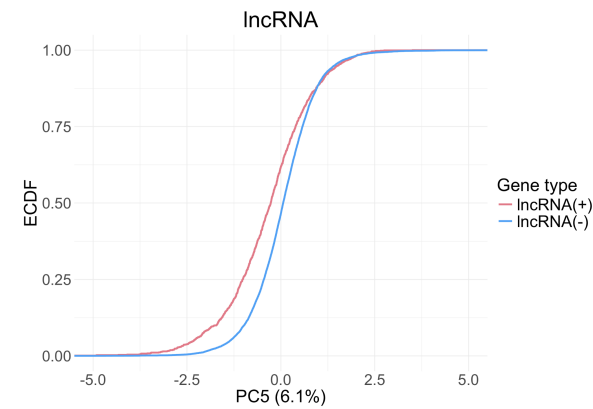
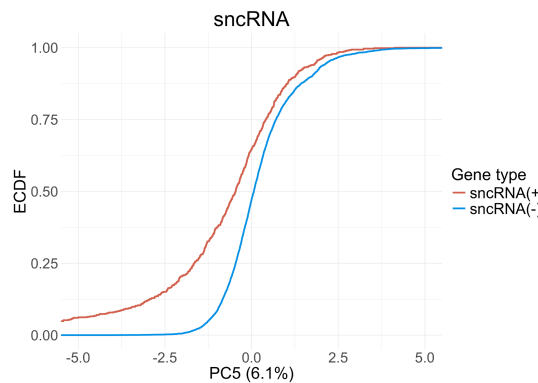
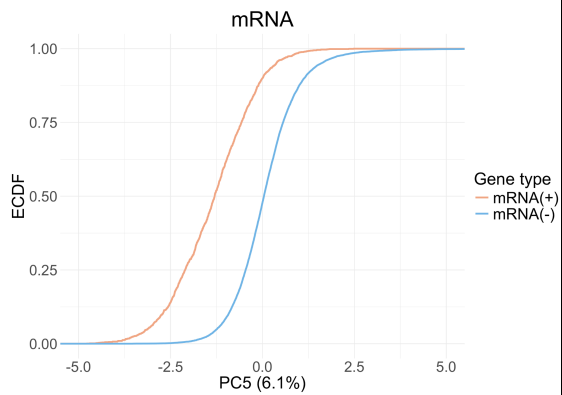
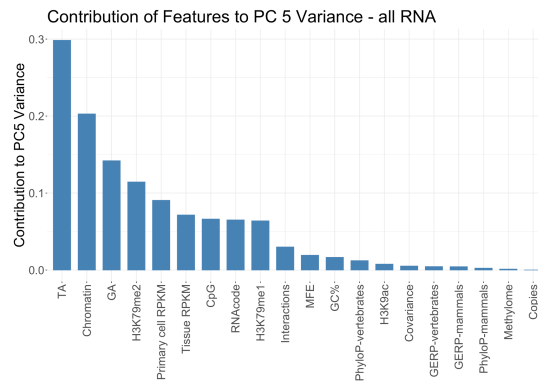
To provide an overall multivariate view of the data, we performed a principal component analysis (PCA) on the robust z-scores of the most informative genomic features. Unlike the per-feature analyses presented in the main text, PCA summarises how features act in combination, allowing us to assess whether functional and non-genic sequences separate in reduced-dimensional space and to identify which variables contribute most strongly to that separation. These plots therefore serve as a complementary overview of the structure, redundancy, and discriminatory power of the selected feature sets across mRNA, sncRNA, and lncRNA datasets.

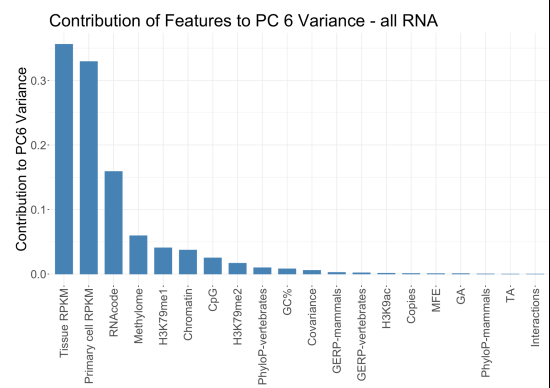
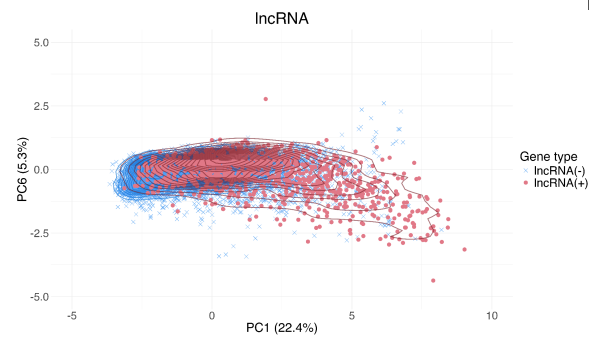
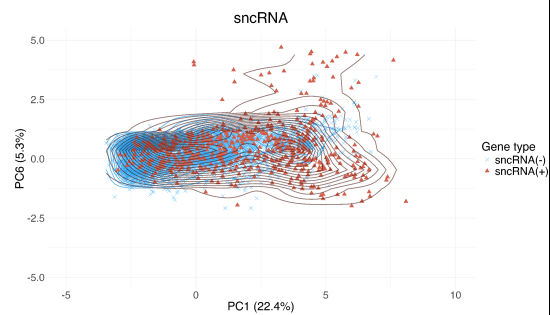
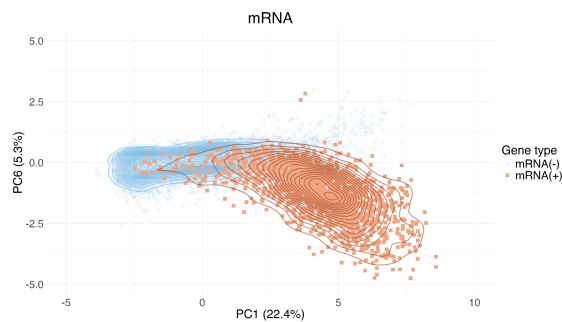
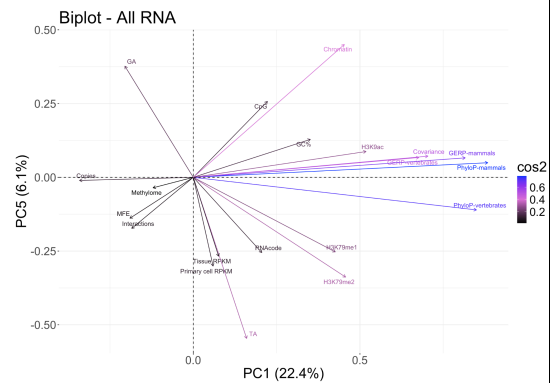












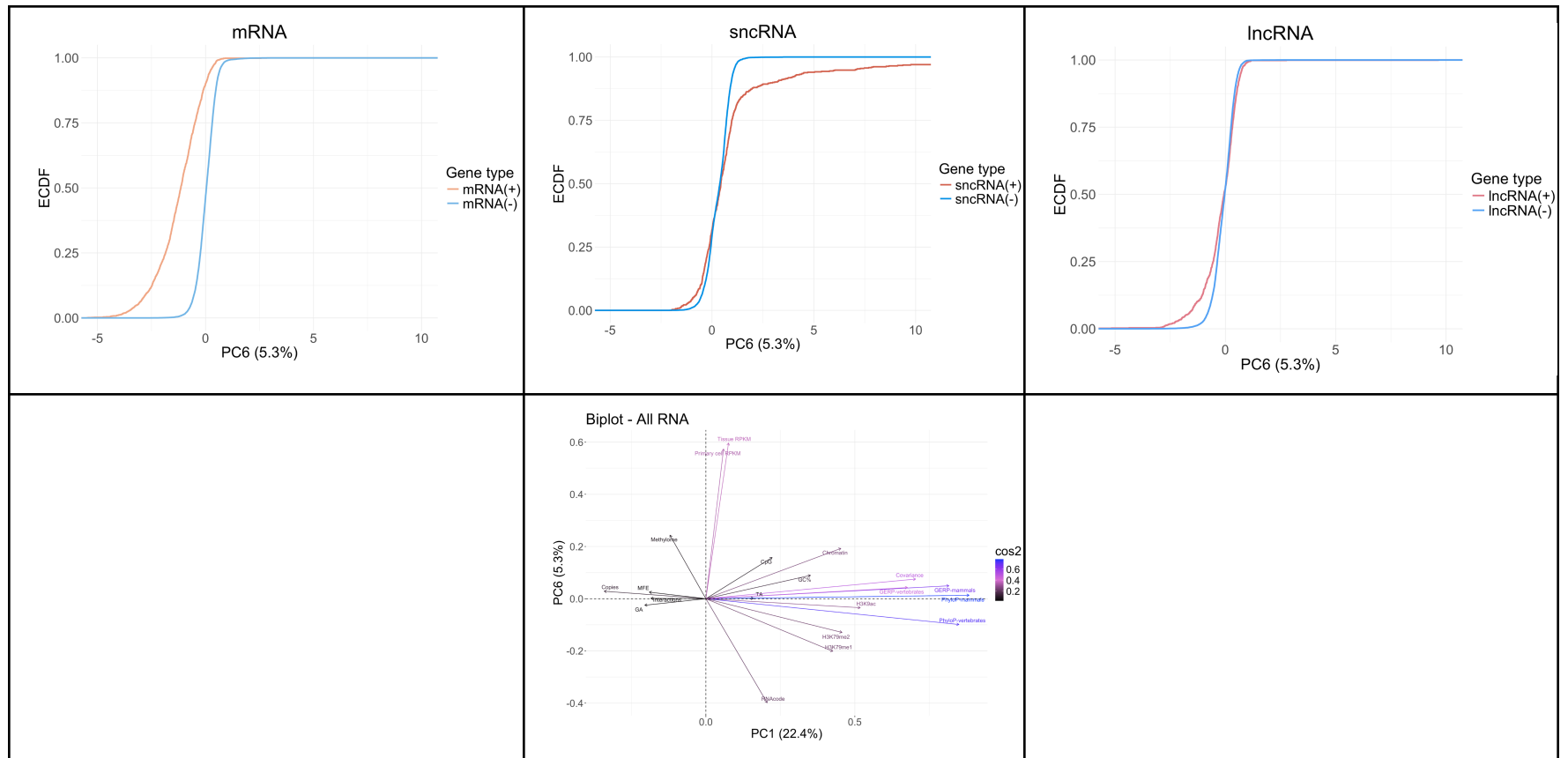


Fig S14. Principal Component Analysis (PCA). The plots include scatter and contour plots, screeplot, univariate plot, and biplot for PC1-PC6. The scatter and contour plots show the clustering of positive and negative sequences for each gene type between PC1 vs PC2-PC6. Screeplot indicates the contribution of each genomic feature to its corresponding PC. The univariate plot illustrates the empirical cumulative distribution function (ECDF) for each PC divided by gene type. Biplots show feature loadings in PC1 vs PC2-PC6 space, vector arrows indicate the direction and magnitude of feature contributions to each principal component.